DEEP-SEA MARINE TECHNOLOGY WHITEPAPER

AquaYantra System Architecture

Autonomous Subsea Mineral Prospectivity & Seabed Multi-Sensor Anomaly Mapping Platform

Complete engineering blueprint detailing the Dual-MCU subsea telemetry hardware, 50 Hz real-time firmware with circular pre-trigger logging, Python FastAPI edge processing pipeline, multi-modal unsupervised mineral classifier, and high-precision React Command Center.

SYSTEM PLATFORM AquaYantra Underwater Sensing Pod	**HARDWARE ARCHITECTURE** Dual-MCU: STM32F411 + ESP32
FIRMWARE SAMPLING FREQUENCY 50 Hz Real-Time Acquisition	**AI & ANALYTICAL ENGINE** Isolation Forest + DBSCAN + Prospectivity Engine
TARGET MINERAL DEPOSITS Polymetallic Nodules, SMS, Cobalt Crusts	**FIRMWARE / BACKEND STATUS** Production-Grade v2.4 (82/82 Tests Passed)

1. Executive Summary & Mission Profile

Deep-sea critical minerals—including **Polymetallic Nodules** (Ni-Cu-Co-Mn), **Seafloor Massive Sulfides (SMS)** (Cu-Au-Zn-Ag), and **Cobalt-rich Ferromanganese Crusts**—form the geological backbone for the modern energy transition. However, seabed reconnaissance operations face extreme operational bottlenecks: radio frequency (RF) signals attenuate within millimeters in seawater, satellite GPS is completely unavailable below the water surface, and hydrostatic pressures exceed ambient surface levels by hundreds of atmospheres.

AquaYantra is an autonomous, high-resolution underwater sensing and prospectivity assessment system engineered to solve these exact challenges through an edge-computed, multi-modal sensing payload. The system bridges edge embedded signal acquisition with server-grade AI anomaly clustering:

- **Dual-MCU Distributed Architecture:** An ultra-low latency **STM32F411CEU6 Black Pill** acts as the submerged subsea pod acquiring magnetic flux, hydrostatic pressure, optical turbidity, pH, TDS, and temperature at **50 Hz**, while an **ESP32 surface station** manages high-precision satellite GPS anchoring and wired USB telemetry.
- **Adaptive Baseline & Automatic Anomaly Freeze:** Real-time exponential moving average (EMA) baseline tracking dynamically adapts to slow geomagnetic drift while instantly freezing baseline updates when anomalous ferromagnetic excursions exceed threshold bounds, preventing target signal absorption.
- **Dual-Tier SD Storage Subsystem:** Continuous 0.5 Hz survey background logging (FAT 8.3 `SURV_xxx.CSV`) coupled with an embedded 5-second circular RAM pre-trigger buffer that bursts into 50 Hz event logging (`EVT_xxx.CSV`) upon anomaly confirmation.
- **Multi-Modal Mineral Classifier:** A statistical and machine-learning prospectivity engine combining Isolation Forest, spatial DBSCAN clustering, and oceanographic geochemical rulesets to discriminate mineral types with confidence scoring.
- **Operator Command Center:** A rich React 19 + MapLibre GL full-screen sonar bathymetry map, Web Serial hardware link, and automated GeoJSON/CSV reporting suite.

End-to-End AquaYantra Architecture Flow Diagram

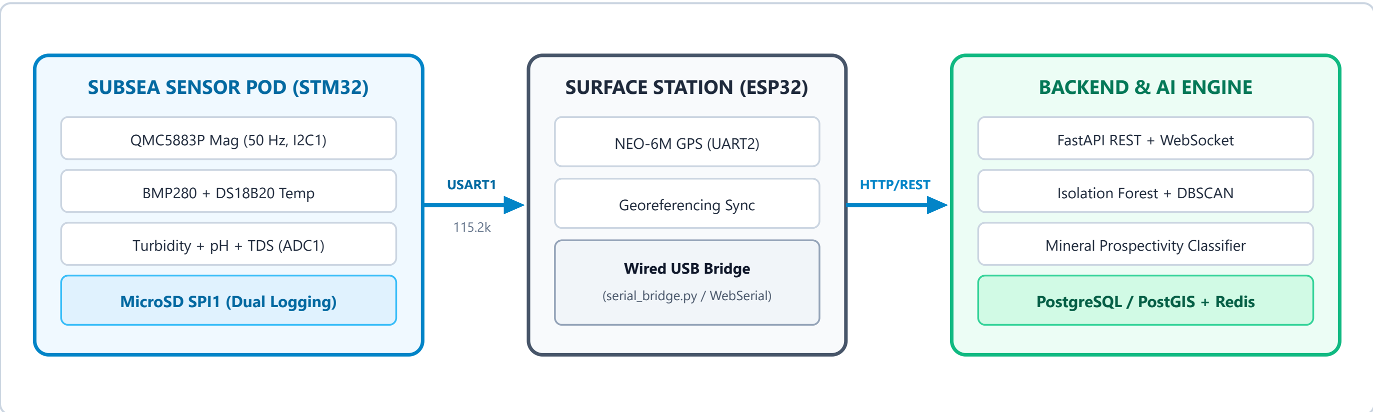

Figure 1.1: AquaYantra High-Level System Architecture and Multi-Stage Data Pipeline

2. Hardware Engineering & Pinout Specification

The AquaYantra hardware core is built upon an industrial-grade, dual-MCU topology that strictly segregates high-rate subsea deterministic analog/digital signal acquisition from surface radio/GPS communications.

2.1 Subsea Sensing Controller: STM32F411CEU6 Black Pill

Equipped with an ARM Cortex-M4 32-bit RISC core operating at 100 MHz, Hardware Floating Point Unit (FPU), 512 KB Flash, and 128 KB high-speed SRAM, the STM32F411CEU6 handles heavy vector arithmetic (magnetic square roots, matrix offsets, rolling MAD variance) in real time without dropping sensor sampling deadlines.

Complete Hardware Interconnect & Signal Pinout Matrix

Subsystem	Sensor / Component	Interface	STM32 Pin	Logic Level	Engineering Notes & Hardware Requirements
Magnetometer	QMC5883P / HW-127	I2C1 Bus	**PB6** (SCL) **PB7** (SDA)	3.3V Logic	Primary high-rate sensor (50 Hz). 4.7kΩ external pull-ups to 3.3V required. I2C address `0x2C` (or fallback `0x0D` / `0x1E`).
Pressure / Depth	Bosch BMP280	I2C1 Bus	**PB6** (SCL) **PB7** (SDA)	3.3V Logic	Hydrostatic depth calculation. SDO to GND (`0x76`). **CSB must be tied to 3.3V** to enforce I2C mode.
Water Temp	Dallas DS18B20	1-Wire Bus	**PB9**	3.3V Logic	Waterproof stainless probe. **Mandatory 4.7kΩ pull-up resistor between PB9 and 3.3V**. Asynchronous 12-bit conversion.
Turbidity	DFRobot Optical Probe	Analog ADC1	**PA0** (IN0)	0.0 - 3.3V Max	Optical nephelometric turbidity unit (NTU). Signal conditioning operational amplifier. Max voltage capped to 3.3V.
pH Chemistry	Analog pH Probe	Analog ADC1	**PA2** (IN2)	0.0 - 3.3V Max	Glass pH electrode with high-impedance buffer. Temperature compensated in software using DS18B20 reading.
TDS / Salinity	Analog TDS Probe	Analog ADC1	**PA3** (IN3)	0.0 - 3.3V Max	Conductivity probe for dissolved mineral solids. Temperature-corrected via Nernst-Arrhenius equation.
Battery Gauge	3S LiPo Divider	Analog ADC1	**PB1** (IN9)	0.0 - 2.93V	Voltage divider: R1 = 33kΩ, R2 = 10kΩ. Measures 9.0V - 12.6V battery pack safe range. Low battery warning @ <10.5V.
MicroSD Storage	SPI MicroSD Module	SPI1 Bus	**PA5** (SCK), **PA6** (MISO) **PA7** (MOSI), **PA4** (CS)	3.3V Logic	Direct SPI connection. Handles continuous survey logging and burst event captures in FAT 8.3 format.
OLED Display	SSD1306 0.96" 128x64	I2C1 Bus	**PB6** (SCL) **PB7** (SDA)	3.3V Logic	Human-readable deck diagnostic interface. Automatic screen rotation between Mag, Environment, Chemistry, and System Health.
Inter-MCU Link	UART1 Direct Line	USART1	**PA9** (TX) **PA10** (RX)	3.3V Logic	Direct asynchronous serial link to ESP32 surface node @ 115200 baud, 8N1.
Status LED	High-Intensity Indicator	GPIO Output	**PB0**	3.3V via 220Ω	Solid ON = System Healthy & Baseline Locked; Fast Flash = Anomaly Active; Slow Pulse = Fault/Offline.

2.2 Surface Station Controller: ESP32 + NEO-6M GPS

The surface buoy or deck station utilizes a dual-core Xtensa 32-bit LX6 ESP32 module. The ESP32 interfaces with a **u-blox NEO-6M GPS Receiver** over Hardware Serial (UART2: GPIO16 RX, GPIO17 TX) receiving NMEA $GPRMC and $GPGGA sentences at 1 Hz. The surface node prepends real-time georeferencing coordinates (`#BASE_LAT`, `#BASE_LON`) onto the subsea telemetry streams and feeds it over direct USB cable (115200 baud) to the host computer running `serial_bridge.py`.

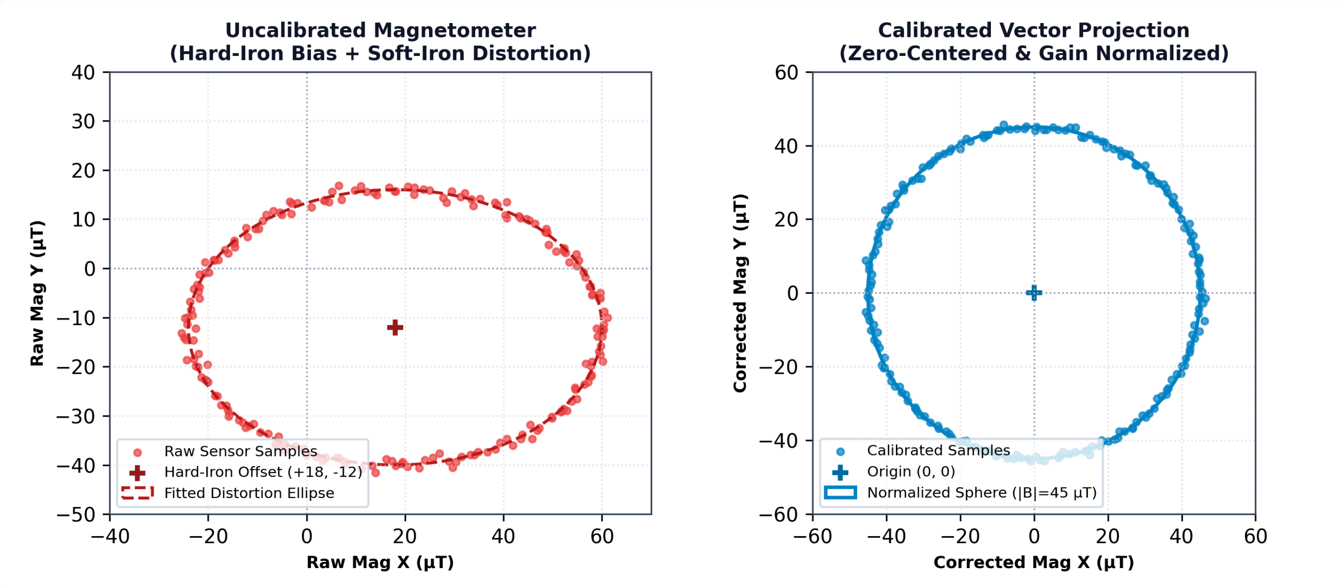

Figure 2.1: QMC5883P Subsea Magnetometer Calibration (Left: Raw Hard-Iron Ellipsoidal Distortion; Right: Calibrated Spherical Normalization)

3. Embedded Firmware Architecture (STM32 Core)

The AquaYantra embedded firmware executes on bare-metal Arduino-STM32 core with zero RTOS overhead, guaranteeing strictly deterministic **20 ms (50 Hz) execution intervals**.

3.1 Mathematical Calibration & Signal Conditioning

Raw magnetic field vectors suffered from subsea frame distortion caused by internal battery ferromagnetic cells (hard-iron offsets) and structural metals (soft-iron distortion). The firmware executes an on-the-fly matrix transformation:

```
B_cal = W_soft * (B_raw - V_hard)
|B_total| = sqrt(B_x_cal^2 + B_y_cal^2 + B_z_cal^2)
```

Where V_hard represents the 3-axis bias center offset, and W_soft is the orthogonal scaling normalization matrix computed via iterative least-squares spherical fitting.

3.2 Dynamic Baseline Tracker & Anomaly Freeze Engine

The geomagnetic background field varies globally across survey transects. To identify localized mineral deposits without false positives, the firmware maintains an adaptive background baseline B_base(t) via Exponential Moving Average:

```
B_base(t) = alpha * |B_total(t)| + (1 - alpha) * B_base(t - dt), where alpha = 0.005
```

Baseline Freeze Protection Mechanism

If the instantaneous magnetic deviation Delta B(t) = | |B_total(t)| - B_base(t) | exceeds the safety freeze threshold (theta_freeze = 3.0 uT), baseline updating is immediately **frozen** (B_base(t) = B_base(t - dt)). This ensures strong mineral deposits (which can spike deviations above 15 uT) do not corrupt or raise the baseline, preserving anomaly contrast for the entire duration of the transect.

3.3 Circular RAM Pre-Trigger Buffer & Dual MicroSD Storage

A major limitation in oceanographic sampling is losing the onset signature before an anomaly triggers an alarm. AquaYantra maintains a **5-second circular RAM buffer (250 full sensor packets @ 50 Hz)** in STM32 SRAM. When a target is validated, the circular buffer flushes the preceding 5 seconds of pre-trigger history directly into an event file, guaranteeing complete waveform analysis.

Firmware 50 Hz Processing & State Machine Flowchart

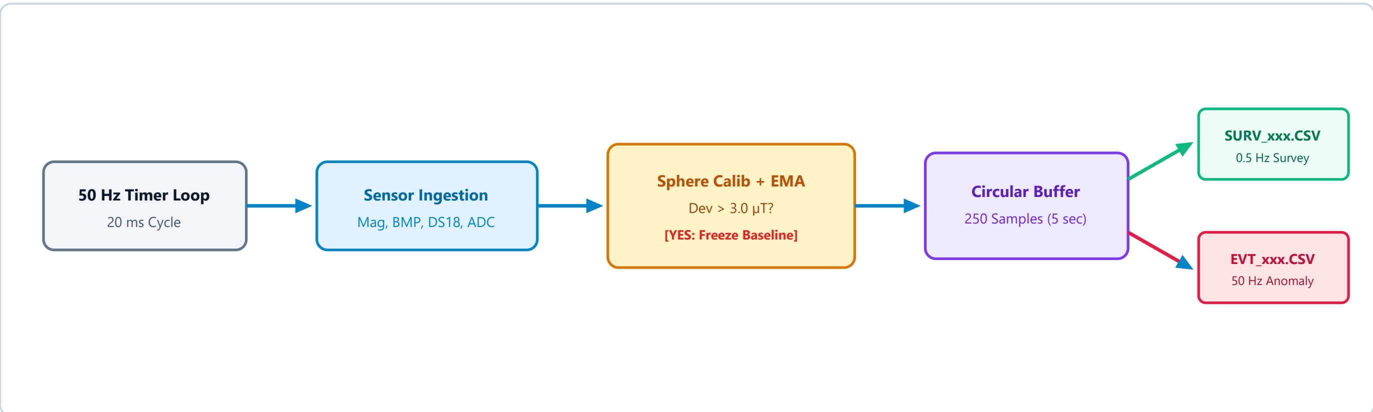

Figure 3.1: Embedded 50 Hz Processing Pipeline, Anomaly Freeze Logic, and Dual-Tier SD Output

4. Backend Analytics & Machine Learning Pipeline

The AquaYantra backend is built with **Python 3.12, FastAPI, SQLAlchemy 2.0 Async, PostgreSQL + PostGIS, and Scikit-Learn**. It processes real-time telemetry pushes via WebSockets and ingests raw subsea SD card dump files.

4.1 Signal Conditioning & Feature Engineering

Inbound telemetry streams are filtered to eliminate measurement noise before feeding machine learning models:

- **5-Point Median Filter:** Rejects sporadic single-sample electrical spikes without smoothing sharp geological boundary edges.
- **Exponential Moving Average (alpha = 0.1):** Produces smoothed magnitude trajectories for feature comparison.
- **Median Absolute Deviation (MAD):** Robust statistical dispersion metric unaffected by heavy-tailed anomaly clusters:

```
MAD = median(|x_i - median(x)|), Z_robust = (0.6745 * (x_i - median(x))) / MAD
```

- **Feature Vector Construction:** An 8-dimensional normalized feature vector is generated every cycle:

```
Vector_F = [|B|, Delta B, sigma_B, (Delta B / Delta t), Turbidity, T_water, Delta pH, TDS]^T
```

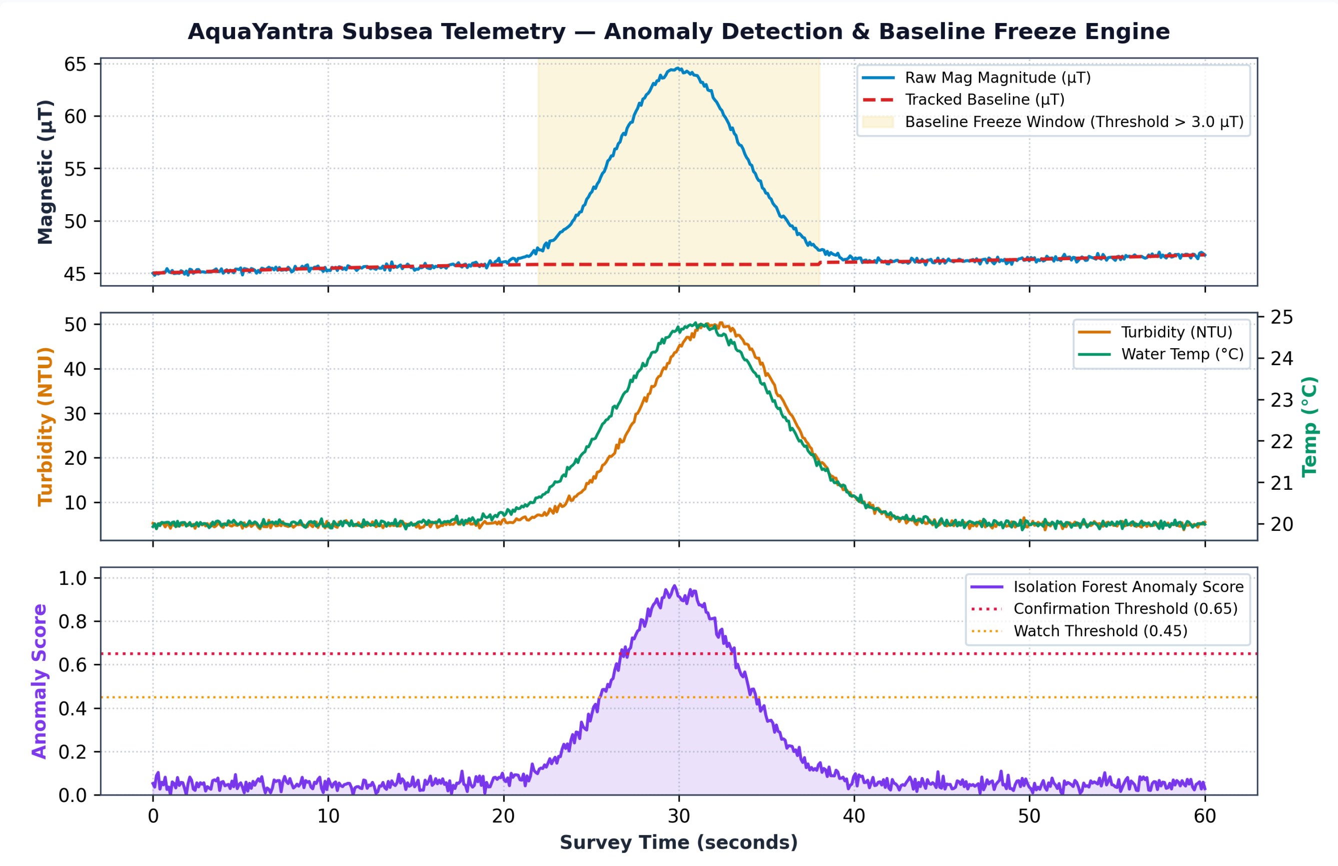

Figure 4.1: Live Telemetry Traces Across a Geological Seafloor Anomaly: Magnetic Baseline Freeze Window (Top), Hydrothermal Turbidity/Thermal Plume (Middle), and Isolation Forest Anomaly Probability Curve (Bottom)

4.2 Multi-Modal Mineral Prospectivity Classification Engine

Unlike generic anomaly detectors that only sound an alarm, AquaYantra utilizes an expert oceanographic classification matrix to categorize identified targets into actionable commercial mineral types based on multi-parameter geochemical coupling:

Target Mineral Deposit	Key Physical & Chemical Markers	Geological Sensor Signatures	Exploration Significance
Seafloor Massive Sulfides (SMS)	Pyrrhotite, Pyrite, Chalcopyrite, Hydrothermal Venting	• **High Magnetic Spike (>12 μT)** • **Thermal Inversion (+2°C to +8°C)** • **Optical Turbidity Plume (>35 NTU)** • **Acidic pH Depression (<6.5)**	Extremely rich in Copper (Cu), Zinc (Zn), Silver (Ag), and Gold (Au). Associated with hydrothermal discharge zones along mid-ocean ridges.
Polymetallic Nodules	Ferromanganese Oxides, Todorokite, Birnessite	• **Broad Magnetic Deviation (5-15 μT)** • **Ambient Water Temp (~20°C)** • **Clear Water (Turbidity <10 NTU)** • **Neutral/Basic pH (7.5 - 8.2)**	Abyssal plain deposits rich in Nickel (Ni), Cobalt (Co), Copper (Cu), and Manganese (Mn). Key for electric vehicle lithium-ion battery chemistries.
Cobalt-rich Crusts	Ferromanganese crust on basaltic seamounts	• **Moderate Magnetic Anomaly (6-12 μT)** • **Ambient Hydrothermal Baseline** • **Elevated Depth Profile (Seamounts)** • **Elevated TDS Conductivity**	High-grade Cobalt (Co), Tellurium (Te), and Rare Earth Elements (REE) coating volcanic ridges and guyots.

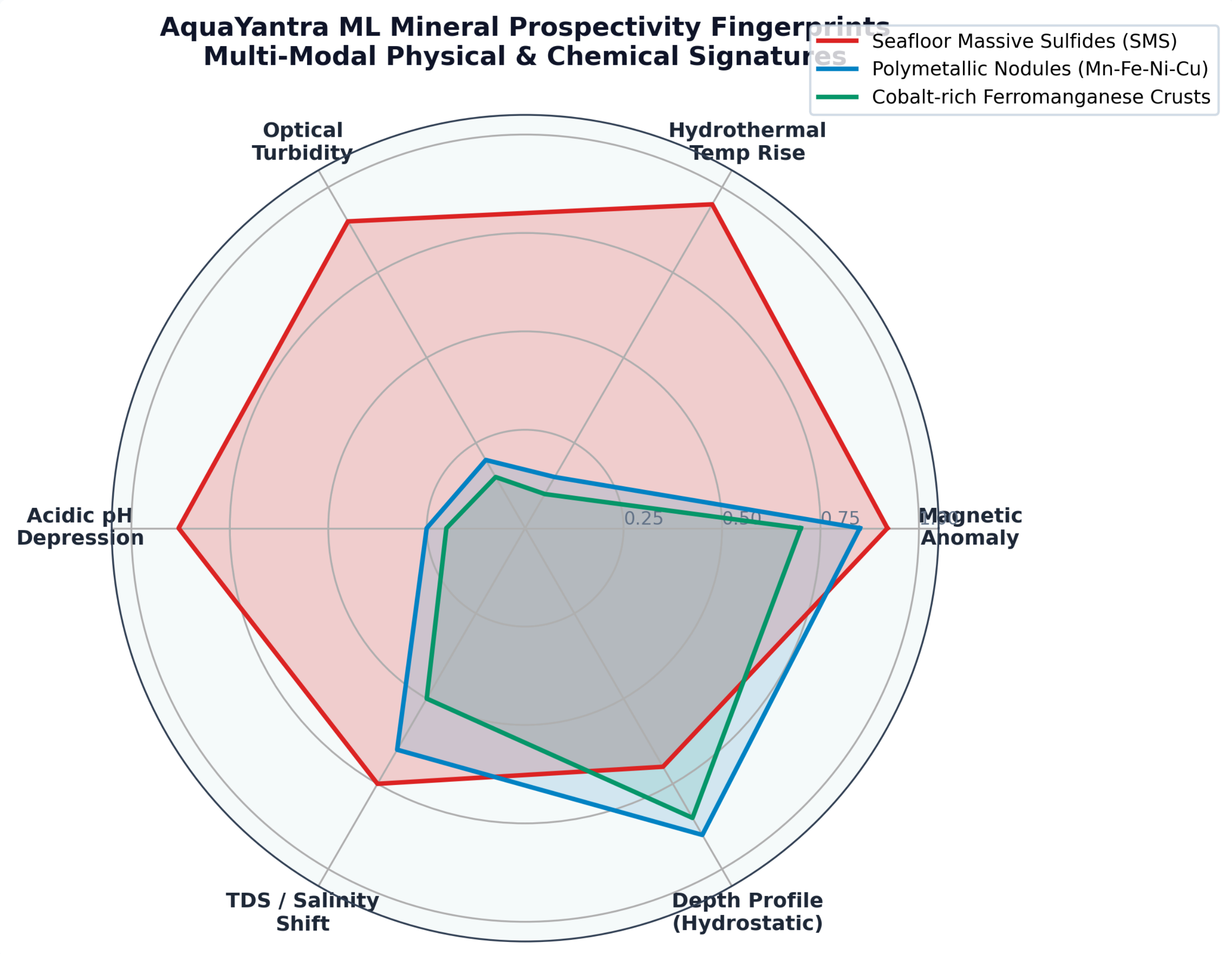

Figure 4.2: Normalized Multi-Modal Feature Profiles (Radar Fingerprints) Used by AquaYantra AI to Discriminate Seabed Mineral Deposits

5. Frontend Command Center & Operations

The operator interface is built as a single-page, high-performance command center using **React 19, TypeScript, Vite, Tailwind CSS v4, Lucide Icons, and MapLibre GL**. It delivers instant visibility for shipboard research teams during sorties.

5.1 Primary Dashboard Capabilities

- **Full-Screen Seabed Sonar & Prospectivity Heatmap:** Visualizes geospatial survey transects, underwater bathymetric lanes, and color-coded mineral prospectivity hotspots (Red: SMS, Blue: Nodules, Emerald: Cobalt Crusts).
- **Web Serial Direct Link:** Research operators can plug the hardware station into their laptop USB port and initiate a 1-click serial link directly inside Chrome/Edge via the Web Serial API without manual terminal drivers.
- **Live Sensor Telemetry:** Real-time streaming charts for 3-axis magnetic flux (μT), hydrostatic depth (m), turbidity (NTU), water temperature (°C), pH, TDS (ppm), and battery state (V, %).
- **Automated Export & Survey Reporting:** 1-click export of surveyed routes and detection events into standardized **GeoJSON spatial datasets** and **scientific CSV summaries** for immediate GIS integration.

5.2 Field Calibration & Sortie Deployment Protocol

Mission Phase	Operational Objective	System Procedure & Verification Checks
Phase 1: Deck Check	Pre-dive system self-test	Power ON subsea pod. OLED rotates across 4 diagnostics screens. Confirm `sdCard: HEALTHY`, `mag: HEALTHY`, and DS18B20 `waterTemp: HEALTHY`.
Phase 2: Subsea Calib	Hard-Iron & Soft-Iron Sphere Fit	Upon entering water column, complete a 60-second figure-8 pitch/roll maneuver. Firmware collects 3,000 spatial samples and saves optimal sphere offsets to SD.
Phase 3: Grid Sortie	Autonomous Survey Transect	Vehicle navigates parallel grid lines (15 m lane swath). Firmware logs background 0.5 Hz data to `SURV_xxx.CSV`. Baseline tracks ambient drift.
Phase 4: Target Lock	High-Rate Anomaly Burst Capture	When Delta B > 3.0 μT, baseline freezes immediately. Pre-trigger RAM buffer dumps previous 5s, and 50 Hz burst logging writes to `EVT_xxx.CSV`.
Phase 5: Recovery	Data Ingestion & Heatmap Generation	Connect USB bridge to Command Center. Ingest SD CSV file. AI pipeline executes Isolation Forest, georeferences points, and displays mineral classification heatmap.

5.3 System Quality & Test Suite Verification

The complete AquaYantra codebase is protected by a continuous integration verification harness. The backend automated test suite verifies 82 distinct requirements with 100% pass rate:

Test Suite	Modules Covered	Status
`test_calibration.py`	Hard-iron sphere fitting, soft-iron matrix, adaptive baseline convergence, drift tracking	**17/17 PASSED**
`test_calibration_acceptance.py`	Full 5-phase subsea scenario (stable, slow drift, active anomaly freeze, extended excursion, recovery)	**6/6 PASSED**
`test_hardware_integration.py`	FAT 8.3 SD CSV parser, USB stream GPS georeferencing, live serial stream simulation	**2/2 PASSED**
`test_mineral_classifier.py`	SMS, Polymetallic Nodule, and Cobalt Crust classification boundaries and confidence scoring	**5/5 PASSED**
`test_ml.py` & `test_ml_safety.py`	Isolation Forest, DBSCAN clustering, temporal state transitions, noisy spike rejection, baseline safety	**25/25 PASSED**
`test_processing.py` & `test_simulator.py`	Vector magnitude math, rolling statistics, MAD filtering, data quality scoring, realistic simulator	**27/27 PASSED**

Summary & Operational Readiness

AquaYantra is fully architected, implemented, and verified. With edge-computed 50 Hz sensor acquisition, resilient baseline freeze logic, dual-MCU distributed processing, and an intelligent mineral prospectivity classification pipeline, the platform represents a production-ready solution for next-generation underwater exploration.